



An online visual measurement method for workpiece dimension based on deep learning

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ABSTRACT

Image preprocessing and edge detection are the two most critical steps when applying machine vision to measure workpiece dimension in industrial field. The surface of the workpiece is usually covered with interference regions, which makes edge detection difficult. Aiming at the problem, the Full Convolutional Network (FCN) model is used to detect interference regions of workpiece images, and then interference regions are processed by the directional texture repair method. To avoid the influence of workpiece surface texture after rough machining, a method based on Holistically-nested Edge Detection (HED) model for rough edge detection of workpiece images is proposed. The edge is obtained by HED, and then it is post-processed based on the non-maximum value suppression and double threshold segmentation methods in the Canny operator to obtain a refined edge image. In order to further improve the accuracy of the dimension measurement, sub-pixel level edge detection accuracy is achieved by cubic spline interpolation. Finally, the method is validated by a shaft workpiece. The measuring accuracy of the outer diameter of the workpiece can reach 0.02 mm, which can effectively meet the requirements of fast semi-finishing inspection. This research provides a common workpiece dimension measurement method in industrial field and methodological guidance for the application of deep learning in industrial inspection.

1. Introduction

In industrial production, quality control determines whether a product meets the delivery requirements, and thus plays an important role along the process of production. Industrial measurement is one of the most important quality control methods, including online measurement and offline measurement. Online measurement superiors to the offline measurement since it has advantages of automation, high reliability, high precision, and low human influence. Among all the types of online measurement, contact measurement is widely used because of its stability and anti-interference ability. However, it takes a long time and affects the processing cycle, and only a few specific points can be measured at a time. With the rapid development of science and technology, non-contact measurement attracts more and more attention due to its wide measuring range, fast measuring speed and low cost.

Commonly used non-contact measurement technologies include laser measurement [1], machine vision [2,3] and so on. In order to get

the profile information of workpiece surface, laser measurement needs to scan the whole surface, which takes a long time. In contrast, the surface information of workpiece can be collected by a camera immediately. The advantages of high precision and high speed makes machine vision more suitable for online measurement than laser measurement.

Traditional machine vision measurement usually detects the edge of workpieces after various image preprocessing methods, and finally transforms the distance between edge pixels into physical distance [4,5] or the surface roughness [6] of workpieces. [4] proposed an on-line machine vision to measure the rubber extrusions, and the measurement uncertainty was constrained within 0.1 mm. [7] proposed a machine visual algorithm and successfully realized the detection of machine tool wear, and the error was controlled 6.7% compared to traditional manual detection. [8] introduced an image-based cross-correlation analysis method to detect the machine tool wear at a microscale down to 0.1 mm. In addition, it has high requirements on the quality of original images and the detection environment, which is the key to

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ensure the success of measurement. However, obtaining high quality images in industrial field is difficult due to the harsh environment. In order to use machine vision technology in industrial field, the following two problems must be solved:

Firstly, there may be interference regions such as cutting fluid and chips on the surface of workpiece. The existence of interference regions will produce noise edges, especially the cutting fluid at the bottom of workpiece affects the integrity of edges.

Secondly, it is difficult to get clear edge image of workpiece by using operator detection or threshold segmentation directly due to the rich texture on the workpiece surface after roughing.

Detection of interference regions is similar to defect detection, which can be achieved by thresholding method. Otsu proposed an automatic thresholding method to solve the unstable problem of fixed threshold [9]. But this method will fail if the histogram is unimodal or close to unimodal. Ng revised the Otsu method for selecting optimal threshold values for both unimodal and bimodal distributions [10]. However, it is still difficult to detect interference regions in the complex processing environment. In addition, it is also difficult to detect the edge with traditional methods. Traditional method has poor robustness and it is also very difficult to adjust parameters [11]. The reason why the traditional method fails in the industrial field is that it cannot distinguish the features to be extracted correctly. Feature engineering plays a very important role in traditional machine vision, and feature selection and extraction are often the core of problem. It is necessary to explore a general online vision measurement method of workpiece dimension, which can automatically select image features.

Compared with traditional methods, the Artificial Intelligence (AI) does not need to perform any preprocessing operations on the input image. It can directly learn the mapping relationship between the input and the target. Research shows that the characteristics of AI make it have good performance in image processing [12]. Support Vector Machine (SVM) [13], Neural Network (NN) [14], and other algorithms are gradually introduced into image processing to replace manual feature extraction.

SVM is mainly applied to classification problems, such as character recognition, face recognition, pedestrian detection, text classification and so on. Therefore, it is not suitable for detecting interference regions and extracting edge images. Though SVM performs well in the case of small samples, its computational cost is high and the calculation speed is slow, and the performance of SVM is gradually surpassed by NN.

The structure of the original NN is simple and difficult to solve the complex image problems. As a kind of NN, Convolutional Neural Networks (CNN) has achieved better prediction results than traditional methods or even other deep network models on many computer vision problems [15]. In the early stage of model implementation, the architecture of CNN is too simple to solve complicated problems. Lecun *et al.* designed LeNet-5 and trained the network model using an error gradient-based algorithm [16,17]. It has achieved great success in handwritten character recognition and laid a solid foundation for the future development of CNN. Since then, deeper and more complex CNN network models such as AlexNet [18], VGG [19], GoogleNet [20], and ResNet [21] have been proposed, and the classification accuracy on ImageNet [22] has been continuously improved. The CNN has entered the stage of extensive research and application. In the past few years, it has been widely used in object detection, semantics segmentation and other image processing fields [23]. In addition, a considerable portion of deep learning focuses on Vision Based Measurement (VBM) systems and CNN are the most widely used deep learning technique, especially in VBM [24].

After the above analysis, CNN is more suitable for solving above mentioned problems of interference detection and edge detection. In addition, there are many specific applications of CNN in industrial detection. Zhou *et al.* proposed a novel end-to-end attention-based fully convolutional neural network framework for automated defect inspection on specular surface, achieves F1 score over 0.7513 (pixel level) and

0.8055 (individual connected components) on the proposed challenging dataset [25]. Huang *et al.* proposed an intelligent Steel Wire Rope (SWR) damage detection method based on CNN [26], which can automatically extract discriminant features by training surface images of the SWR and achieves a F1 score of 0.99. Zhong *et al.* proposed a three-stage automatic defect inspection system for highspeed railways based on CNN [27]. Zhang *et al.* proposed a vision-based fusion method for defect detection of milling cutter spiral cutting edge, and the precision was 98.86% [28]. These application cases show the feasibility of applying CNN to industrial measurement.

The contributions of this paper are as follows:

- (1) FCN deep learning model is used to detect interference regions of workpiece images, and then interference regions are processed by the directional texture repair method.
- (2) To avoid the influence of workpiece surface texture after rough machining, a method based on HED deep learning model for rough edge detection of workpiece images is proposed.
- (3) The measurement accuracy of the proposed method can reach 0.02 mm, which can effectively meet the requirements of fast semi-finishing inspection.

Thus, an online measurement method that utilizes CNN is proposed in this paper to solve the problems of interference detection and edge detection in industrial field. The rest of this paper is organized as follows. Section 2 introduces the general idea of the proposed online measurement method. Section 3 designs the detection and processing method for interference regions detection using FCN model. Section 4 designs the edge detection method using HED model. Then, this method has been implemented and verified through an experiment of shaft detection in Section 5. Finally, conclusions are discussed in Section 6.

2. Overview

It can be seen that applying CNN to industrial measurement is feasible, while for specific problems in the industrial measurement, new models are derived from CNN.

In the field of semantic segmentation, the Full Convolutional Network (FCN) proposed by Jonathan Long has brought about significant changes [29]. This network changes the architecture of CNN, realizes the mapping of image to image. Semantic information refers to the content of image, including image texture, color, shape and so on. Image semantics segmentation refers to annotating different information in image from the perspective of pixels. The process of interference regions detection is to segment interference regions from the background of the workpiece images accurately. FCN can achieve better end-to-end training and image semantics segmentation. It can input the original image, classify each pixel, and output a labeled semantics segmentation image. the FCN model has the following characteristics: (1) It has no limitation on the size of the input image and does not need any preprocessing operations. (2) It classifies each pixel and has scale and rotation invariance. (3) the model architecture has the ability to learn multi-level features. In addition, there are many successful applications based on FCN. Zhang *et al.* proposed a concrete crack detection method based on improved FCN [30], which distinguishes background from crack at the pixel level effectively. Zhou *et al.* propose an approach of combining fully convolutional network and conditional random field for image semantic segmentation [31], which achieves improved accuracy compared with several other methods on Pascal VOC 2012 dataset. According to the above successful applications and characteristics, a preliminary conclusion can be drawn that the model is suitable for interference regions detection.

On this basis, the research of CNN in semantics segmentation is also flourishing [32–34]. Based on the idea of end-to-end of FCN, Xie *et al.* designed a Holistically-nested Edge Detection (HED) model [35]. The model overcomes the shortcomings of traditional edge detection

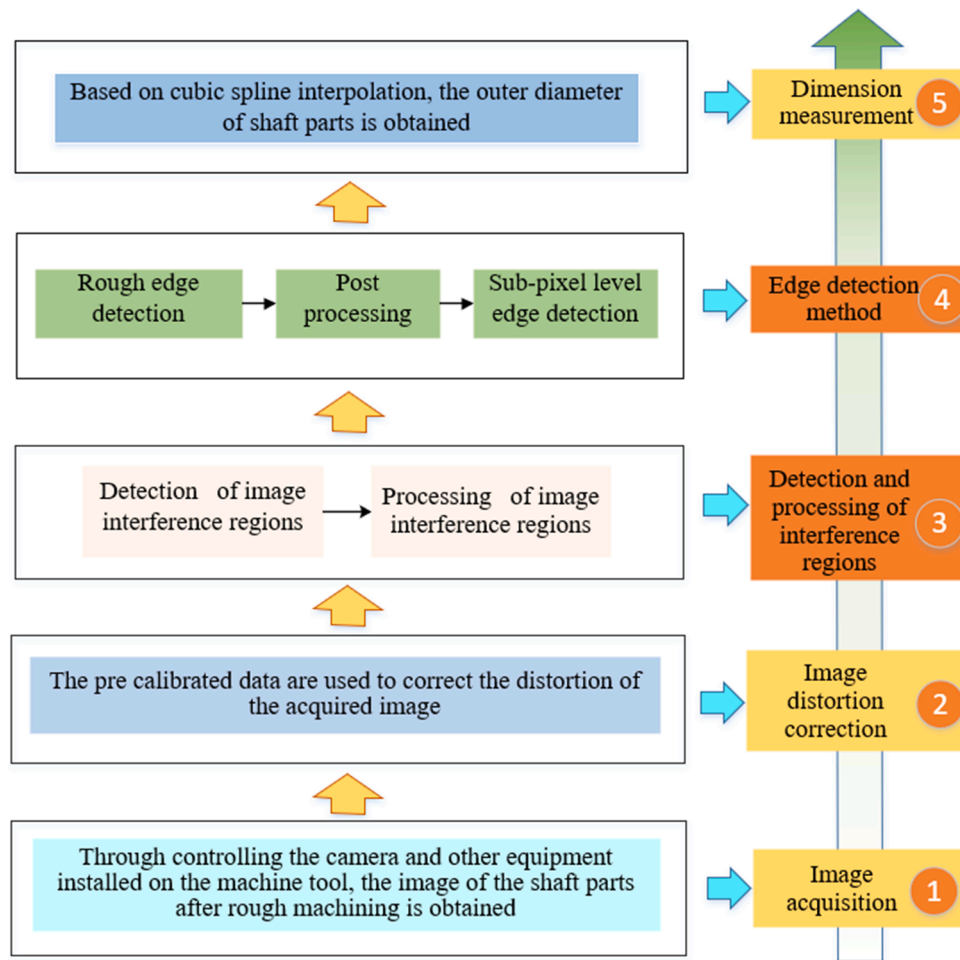


Fig. 1. Flow chart of online visual measurement method based on deep learning.

methods, such as difficulty in parameter adjustment and poor robustness to noise. The edge obtained by HED model is complete, and the robustness to noise is strong. The model is also suitable for edge detection.

However, FCN and HED cannot be directly used in the field of industrial measurement because they are general algorithms, which need to be optimized and improved for specific purpose. In addition, some auxiliary operations are also required to fit the application scenarios. Many factors should be taken into consideration such as network structure, layer number, learning rate, etc. Besides, it is necessary to determine a suitable termination conditions of iteration for the ideal result in practice.

Under the guidance of theory and practice, the original neural network was improved and added some auxiliary operations in this paper to improve the accuracy. Our solutions to measure workpiece dimension are summarized in Fig. 1.

Firstly, the workpiece images are captured and the distortion correction is carried out. Then, the interference region is detected and processed before extracting edge image. Finally, the diameter of the outer circle is obtained through interpolation. The step 3 and step 4 are the focus of this paper, their general processes are introduced next and the implementation methods will be introduced in Section 3 and Section 4 subsequently.

At the step 3, aiming at the problem of interference regions on workpiece surface in the process of machining, the detection of interference regions was studied. In the actual processing environment, 500 workpiece images were collected for training FCN model. In addition, because the detection of interference regions and edge detection are

relatively simple, and the use of fine-tuning, these data sets can meet the preliminary needs through experiments. After labeling, the FCN model that can be used to correctly detect interference regions, was trained with original images as input and corresponding labeled images as output. Finally, the directional texture restoration method was used to restore the interference regions for subsequent edge detection.

At the step 4, aiming at the problem of rich texture on the surface of workpiece, a general edge detection method was explored in this paper. Similar to the FCN model training, HED model was trained with original images as input and the corresponding edge images as output. After workpiece edge is detected using HED model, it is post-processed based on the non-maximum suppression and double threshold segmentation methods in the Canny operator to obtain a refined edge image. Finally, in order to improve the accuracy of dimension measurement, the sub-pixel edge algorithm is used to precisely locate the edge.

3. Detection and processing method for interference regions of workpiece images

In the process of machining, workpiece surface is covered with irregular interference regions such as cutting fluid and chips. To improve the quality of edge detection, interference regions must be detected and processed firstly.

3.1. Detection of interference regions of workpiece images based on FCN

According to the analysis in the introduction, it is feasible to use FCN network to solve the problem of interference regions detection.

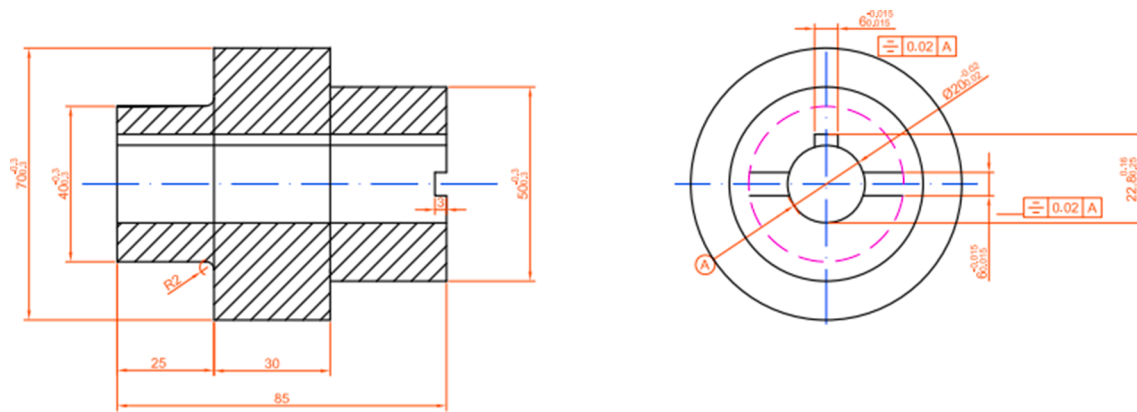
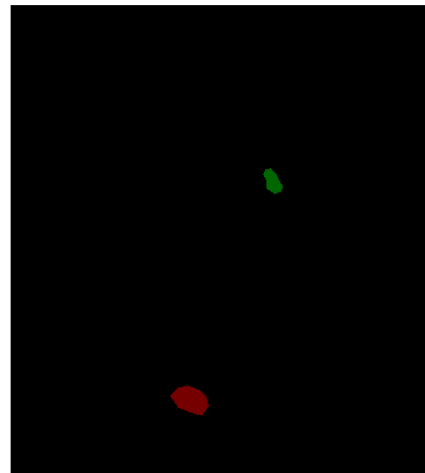


Fig. 2. The design drawing of the shaft workpiece.



(a) the input image of the FCN model.



(b) the label image of the FCN model

Fig. 3. The workpiece image dataset of the FCN model.

Therefore, a method of interference regions detection based on FCN model was proposed.

3.1.1. Architecture construction of FCN model

FCN has a skip architecture, which fuses the feature hierarchy through combining deep semantic information and shallow appearance information. The model consists of seven convolution layers and five pooling layers. The pooling operation will reduce the feature map size to half of the original size, and the convolution operation does not change the feature map size. After five convolution and pooling operations, the size of the output feature map is reduced to $1/32$ of the original image. In order to restore the size of the original image, an up-sampled operation at stride 32 is performed on the output feature map, and each pixel is classified to obtain a network model FCN-32s. Similarly, combining predictions from both the final layer and the pool4 layer, the FCN-16s model can be obtained at stride 16; and combining all predictions from pool3, pool4 and final layer, the FCN-8s model can be obtained at stride 8.

3.1.2. Construction of image dataset used for training FCN

The images of a shaft during processing were taken by industrial cameras at different times in the same environment and then constituted dataset. The design drawing of the shaft is shown in Fig. 2. There are 500 workpiece images in the dataset, and 450 images were selected randomly as training set of FCN, and the remaining 50 images were used as test set of trained FCN model.

Cutting fluid and chips exist in the interference regions of the images. MIT image labeling tool LabelMe [36] is used to improve the efficiency of labeling. After labeling, the original image will be converted into the corresponding labeled image. As shown in the Fig. 3, each closed curve represents an interference region, the red label is cutting fluid and the green label is a chip.

3.1.3. Model training of FCN

Fine-tuning refers to loading the weights of pre-trained model into current model, and then using private dataset to retrain the model, so as to avoid overfitting and improve training efficiency. Through analyzing the logic of FCN, the model has been trained with the idea of fine-tuning. The whole training process of the model is shown in Fig. 4. The FCN-32s model that initialized by the weights of VGG16 [19] trained on ImageNet [22] was trained firstly, then load the weights of the FCN-32s into the FCN-16s and continue training, and finally load the weights of the FCN-16s into the FCN-8s and finish training. At each stage, the model is trained end-to-end and initialized with the weights of earlier model.

The whole process of training can be divided into 3 steps, which are introduced as follows.

Step 1: Initialize and train FCN-32s model.

The VGG16 model trained on the ImageNet [22] was used as the pre-trained model. For FCN-32s construction, the original weights are kept unchanged and all fully connected layers are replaced with convolution layers. Then, FCN-32s model can be trained based on the constructed dataset.

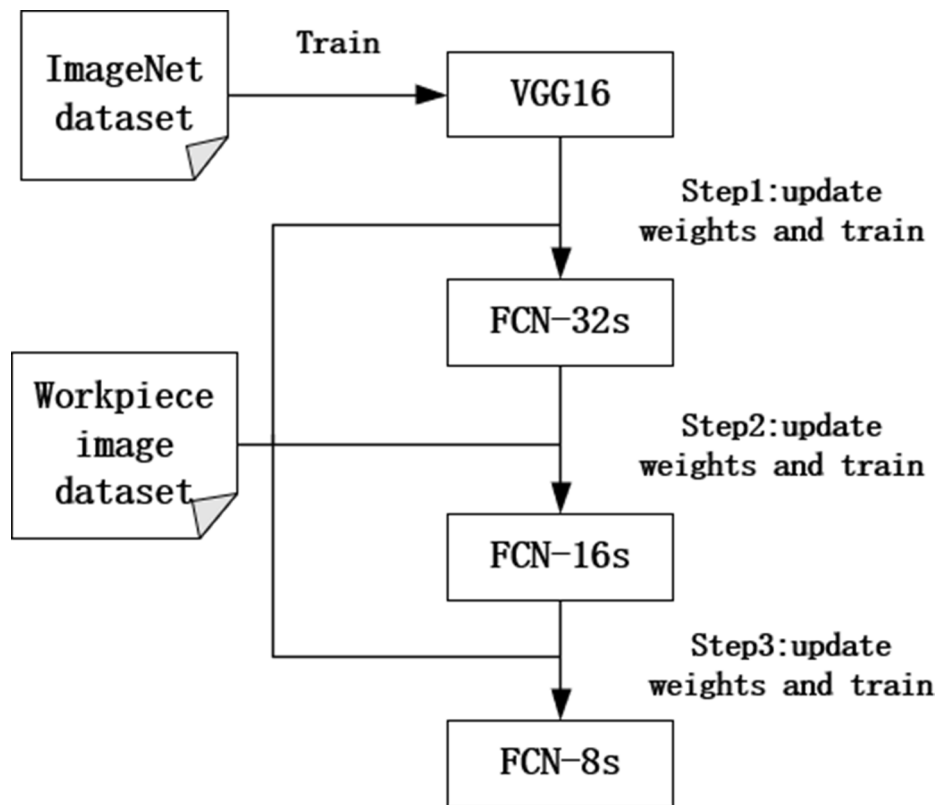


Fig. 4. Training process of the FCN model.

Table 1

Training parameters for FCN.

Loss function	Back propagation parameter	Learning strategy	Learning rate of FCN-32s	Learning rate of FCN-16s	Learning rate of FCN-8s
Softmax function	SGD optimization algorithm	Fixed	1e-10	1e-12	1e-14

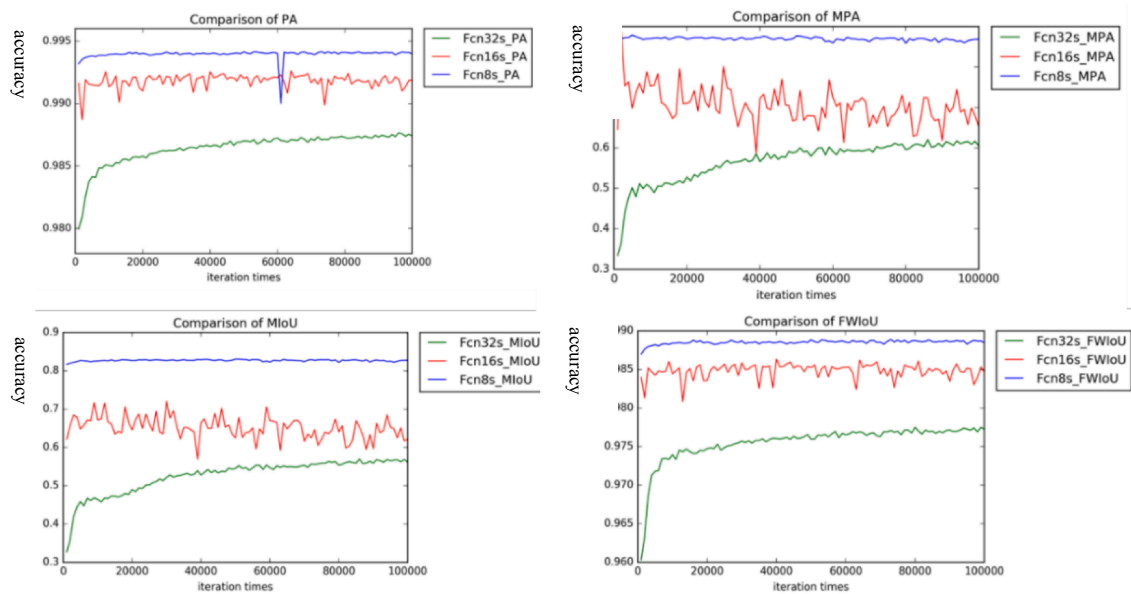


Fig. 5. The evaluation indicators of FCN-32s, FCN-16s and FCN-8s.

Table 2

The evaluation indicators of FCN-32s, FCN-16s and FCN-8s with 5000 epochs.

Models	PA	MPA	MIoU	FWIoU
FCN-32s	0.9872	0.6038	0.5763	0.9761
FCN-16s	0.9921	0.7024	0.6531	0.9851
FCN-8s	0.9940	0.8685	0.8271	0.9884

Step 2: Initialize and train FCN-16s model.

According to the architecture of FCN, it is not necessary to reconstruct the convolution layer, so the weights in FCN-32s can be directly updated to FCN-16s. Then, train FCN-16s with the constructed dataset.

Step 3: Initialize and train FCN-8s model.

Similar to Step 2, update the weights of FCN-16s to FCN-8s and train FCN-8s with the constructed dataset.

As shown in Table 1, these training parameters achieved the best results after many experiments.

3.1.4. Model assessment of FCN

After the model training is completed, the performance of the FCN model needs to be evaluated. The image semantic segmentation field often uses these four evaluation indicators to measure the algorithm [37]: Pixel Accuracy (PA), Mean Pixel Accuracy (MPA), Mean Intersection over Union (MIoU), Frequency Weighted Intersection over Union (FWIoU). As shown in the Fig. 5.

MPA is the standard measure indicator of image semantics segmentation, but FWIoU is more convincing because the amount of cutting fluid is much larger than the amount of chips in the selected images. After analysis, FCN-8s achieves the best performance when iteration times is 50000, therefore it was chosen as the final model. In addition, the batch size of model training is 45, so 10 iterations is an epoch. The evaluation indicators of each model with 5000 epochs are shown in Table 2.

As shown in the Fig. 6, FCN-8s model also achieves the best performance from the illustration intuitively. With the continuous forward integration of network models, the detected interference regions are more standardized and more accurate.

3.2. Processing of interference regions of workpiece images

Interference regions will affect the edge detection of workpiece, especially the cutting fluid accumulate in the workpiece contour easily, which destroys the contour integrity of the workpiece images.

Image inpainting is not the focus of this paper because the performances of existing image inpainting methods can meet the requirements. Telea *et al.* proposed an image inpainting technique based on the Fast Marching Method(FMM) [38], which produces nearly identical results to more complex, and usually slower known methods. Sapiro *et al.* proposed an approach based on the Navier-Stokes(NS) equations for fluid dynamics [39], which has the immediate advantage of well-developed theoretical and numerical results. Wang *et al.* proposed a directional texture image inpainting algorithm, which can repair directional texture image well [40].

As shown in the Fig. 7, for the kind of workpiece images with very rich surface texture, the FMM and NS image inpainting algorithms have not achieved good results after the experiment. The image inpainting algorithm proposed by Wang has a very good performance. Therefore, it was selected to process interference regions of workpiece images.

4. Edge detection method for workpiece images

Through the previous section, the irregular interference regions have been repaired. The repaired image eliminates the influence of interference regions and can be used for edge detection. After edge detection, the diameter of the shaft can be fitted.

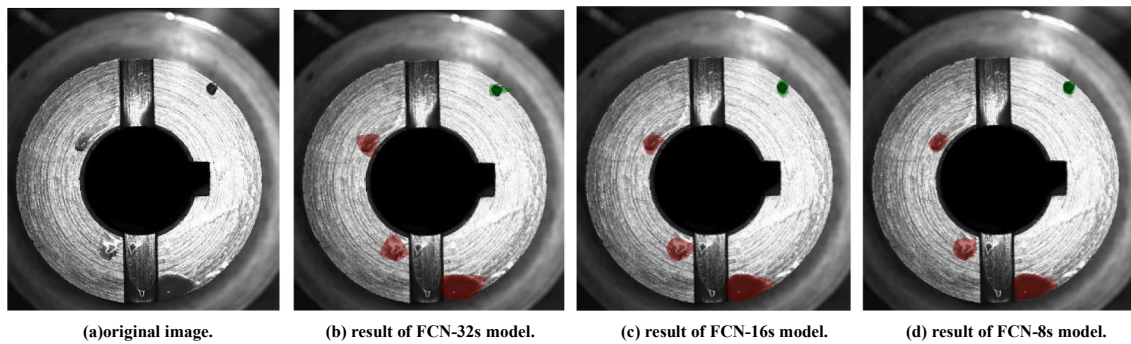


Fig. 6. Detection results of different FCN models.

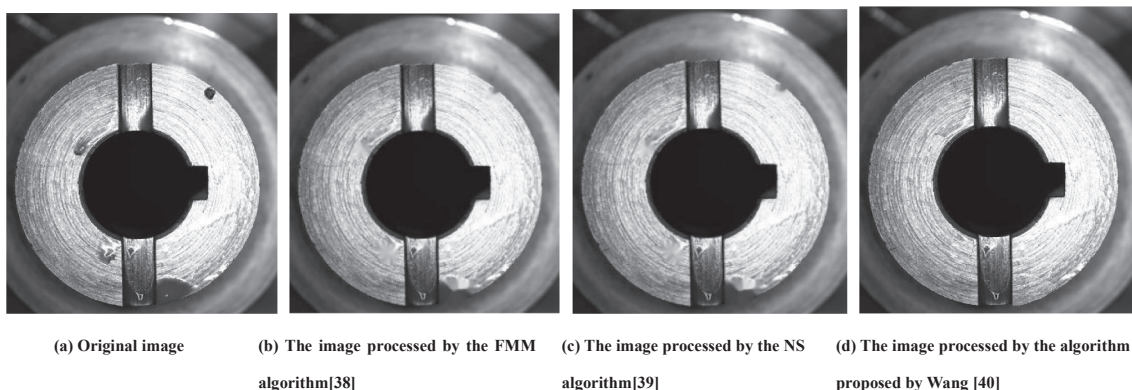


Fig. 7. Processing results of different image inpainting algorithms.

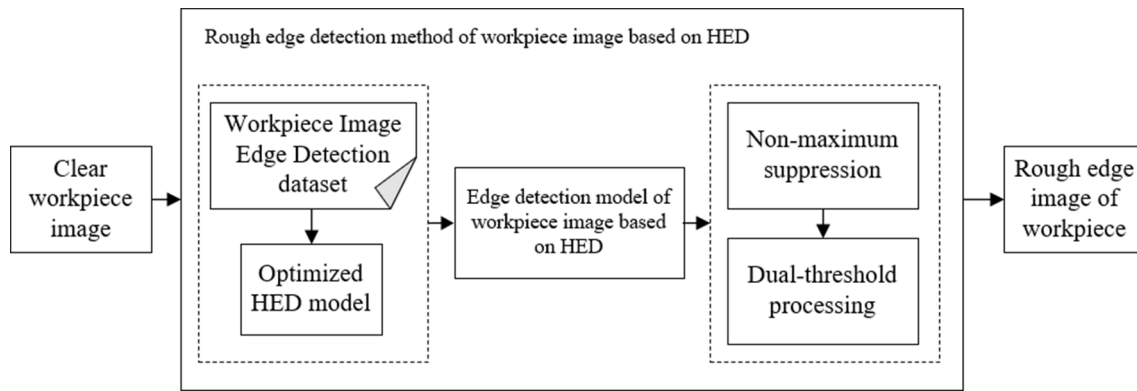
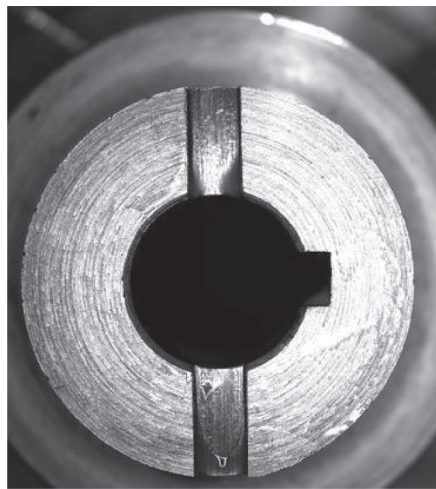
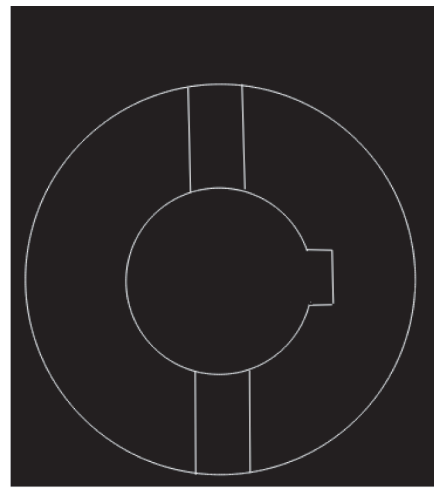


Fig. 8. Rough edge detection method of workpiece images based on HED.



(a) the input image of the HED model.



(b) the label image of the HED model.

Fig. 9. The workpiece image dataset of the HED model.

4.1. Rough edge detection of workpiece images based on HED

HED model is a general edge detection model improved on the basis of FCN and VGG model [35]. The edge obtained by HED model is complete, and its robustness to noise is strong. This model has reached the human level in the standard benchmark test.

However, there is still a problem of edge blurring when HED model is used to extract the edge of the workpiece from images. To solve this problem, the following two improvement methods are proposed. Firstly, improve the training method of HED model. In the process of backward propagation of the traditional training, the feature maps output at each scale are involved in the calculation of the loss value. While in this paper, only the fused images are used to calculate the loss values. Secondly, the edge images that output by the modified HED model are further processed by the non-maximum suppression and double threshold segmentation algorithms of Canny operator.

The flow of rough edge detection is as follows (see Fig. 8).

4.1.1. Construction of image dataset used for training HED

After interference region processing, the repaired images constituted the dataset, which are used for training HED model. 500 workpiece images after processing interference regions are used as dataset. 450 images were selected as training set of HED model at random, and the remaining 50 images were used as test set of HED model after training. For the application scenario of industrial measurement, it is necessary to make the edge image of the workpiece as a class label. As shown in

Table 3

Training parameters for improved HED.

Loss function	Back propagation parameter	Learning strategy	Learning rate of FCN-32s	Learning rate of FCN-16s	Learning rate of FCN-8s
Weighted cross entropy function	SGD optimization algorithm	Fixed	1e-10	1e-12	1e-14

Fig. 9, the model takes processed workpiece images as input, and outputs the hand-labeled edge of the corresponding workpiece images.

4.1.2. Characteristics and training of improved HED

HED is a general edge detection model and owns a multi-scale and multi-level network architecture. It can achieve end-to-end training.

In the scene of original paper [35], images are usually complex with many edges to be detected. If only the loss value of the last layer is calculated for training, some edges will not be detected due to the lack of deep information. The global nesting architecture takes all levels of information into account to ensure the image quality. While for the scene of industrial measurement, images are usually simple with a few regular edges of workpiece. And the global nesting architecture will blur the edge because it takes too much deep information into account. In order to make HED more suitable for industrial measurement, only the fused images are used to calculate the loss values in the paper. Compared with

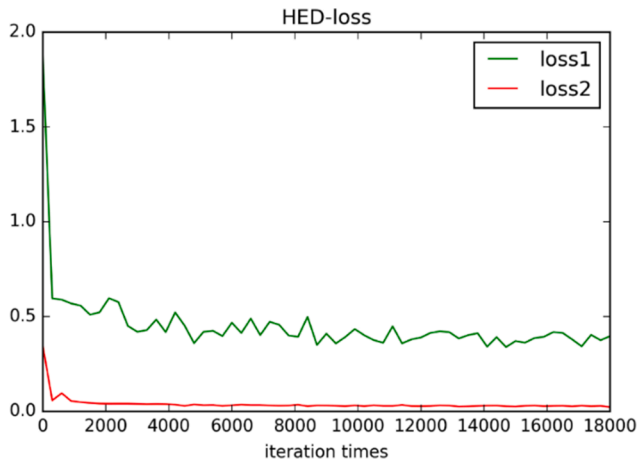


Fig. 10. The loss functions of HED models. (loss1 represents the loss function value of the original HED model. loss2 represents the result after optimizing.)

the original HED model, the edge of the workpiece images detected by the improved model is finer and the location is more accurate.

As shown in Table 3, these training parameters achieved the best results after many experiments.

4.1.3. Model assessment of improved HED

As shown in the Fig. 10, the value of loss function decreases and converges gradually, indicating that the edge of workpiece images generated by HED model approximates the edge of the labeled image gradually.

According to Fig. 10, the loss value decreases before 9000 iteration times and fluctuates steadily after 9000 iteration times. In other words, the model with 900 epochs is chosen as the final detection model. As shown in the Fig. 11, It can be seen that the edge of the workpiece image detected by the improved HED model is more accurate and clearer.

4.1.4. Image post-processing

The edge detection model of workpiece images based on HED can obtain complete edges and less interference contours. To further improve the accuracy of the edge, the non-maximum suppression and double threshold segmentation in Canny operator detection are adopted.

(1) Non-maximum suppression

From the perspective of pixel gray, the edge of workpiece is the place where the gray value of the image pixel changes the most. Non-maximum suppression aims at suppressing the points where the gradient of the pixels in the local region is not the maximum, so that some non-edge points can be removed. The edges obtained by non-maximum suppression method are further refined, but there are still some false edges.

(2) Double threshold segmentation

Double threshold segmentation algorithm was used to further detect and connect edges of the image. In this paper, the threshold obtained by Otsu algorithm [9] is used as the high threshold T_1 , and the $1/2$ of T_1 is used as the low threshold T_2 . The pixel type can be obtained by the formula (1).

$$pixel = \begin{cases} \text{strong edge pixel} & t > T_1 \\ \text{weak edge pixel} & T_1 > t > T_2 \\ \text{suppressed edge pixel} & t < T_2 \end{cases} \quad (1)$$

All edges in a strong edge image are recognized as real edges. For the pixels in a weak edge image, they are retained if there are strong edge pixels in its eight neighborhoods. In this way, the broken strong edges

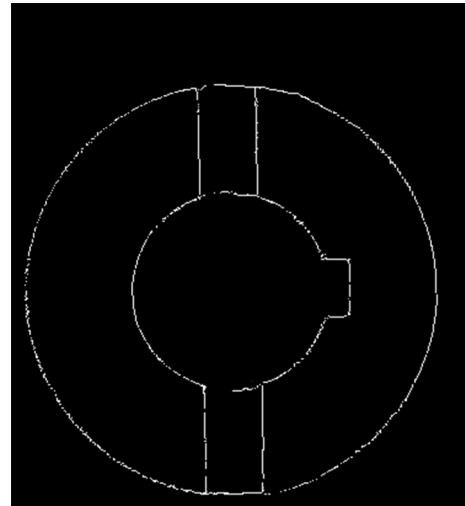
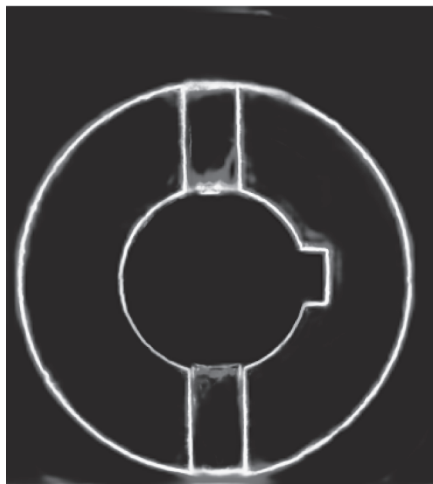
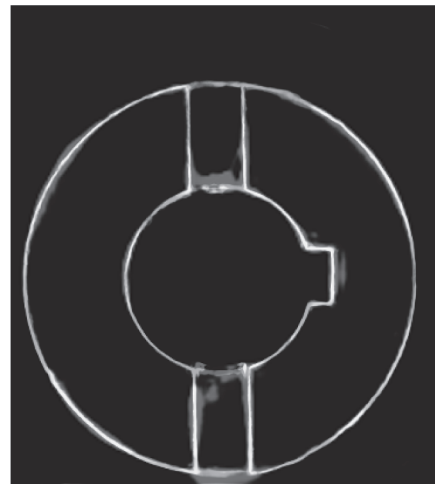


Fig. 12. Edge detection result after post-processing.



(a) the result of the original HED model.



(b) the result after optimizing.

Fig. 11. Edge detection results of workpiece images.

Table 4

X-direction sub-pixel coordinate.

Number of pixels	Rough edge ordinate	Rough edge abscissa	Sub-pixel abscissa
1	50	130	129.067612
2	50	338	338.034850

Table 5

Y-direction sub-pixel coordinate.

Number of pixels	Rough edge abscissa	Rough edge ordinate	Sub-pixel ordinate
1	100	81	80.060291
2	100	153	152.461962
3	100	200	199.149723
4	100	278	277.51657

can be connected. As shown in Fig. 12, the rough edge image of the workpiece is more accurate and clearer after post-processing.

4.2. Sub-pixel edge detection of workpiece images

In industrial measurement, it is necessary to precisely locate sub-pixel edges because edge localization at the pixel level cannot meet the accuracy requirements. In this paper, the cubic spline interpolation [41] method is used to obtain sub-pixel edges, which has characters of simple calculation and high accuracy. The edge pixels of the image after rough edge detection were used to calculate sub-pixel edges. For an edge pixel, its subpixels in X-direction can be obtained by selecting some pixels in X-direction for cubic spline interpolation. It's the same thing for Y-direction. Finally, the sub-pixels of the image are calculated.

Some sub-pixel interpolation data were listed. The abscissa edge is accurately positioned with the ordinate of 50, and the corresponding data is shown in Table 4. While the ordinate edge is accurately positioned with the abscissa of 100, and the corresponding data is shown in Table 5.

So far, an accurate sub-pixel edge image of workpieces has been obtained. Next, the obtained sub-pixel edge coordinates were fitted by the least square method to calculate the workpiece outer circle diameter.

5. Measurement implementation and verification

A shaft in the practical industrial production environment was measured as a case. Firstly, the experiment design is introduced, and then the implementation method of measurement is explained. Finally, the measurement results of the shaft are analyzed.

5.1. Experimental design

In order to verify the effectiveness of the method, the results of coordinate measuring machine were selected as the benchmark and compared with the measurement results from experiment.

Components of the image acquisition system were selected and installed in the CNC lathe to build a hardware platform. Industrial camera model is MV-CA050-10GM of HIKVISION, lens model is MF2520M-5MP of HIKVISION and light source is LED annular light.

In order to ensure the image quality, the camera system and the light source system are required to be coaxial with the workpiece. Therefore, the camera bracket was designed according to the tailstock structure, which can be directly embedded in the tailstock of CNC machine tool. The light source can be directly coaxially mounted on the lens. The relative position of the industrial camera, the workpiece and the light source are shown in Fig. 13.

5.2. The implementation of measurement

The proposed methods are applied to online measurement of shaft. After the rough machining of workpieces, the vision system will collect the image and give the outer circle dimension to determine whether the workpiece dimension meets the requirements of the next semi-finishing. The workpiece images are collected in the CNC lathe processing of a shaft.

Camera calibration is an indispensable part of online visual measurement of workpiece dimension, which should be done before image collecting. The measurement system can obtain internal parameters by calibration and correct the distortion caused by manufacturing process to reduce the influence of image distortion on dimensional measurement accuracy. The internal parameters of the camera are obtained by OpenCV, and the distortion of the image is corrected. Then the physical space is calibrated by high precision standard blocks to calculate the pixel equivalent, which can be obtained from the following equation:

$$k = \frac{a}{r} \quad (2)$$

where a is the actual dimension of the workpiece and r is the number of pixels. The unit of pixel equivalent k is mm/pixel. Its physical meaning is

Table 6

Calibration results of a standard block of 30 mm.

Number of measurements	Physical dimensions	Number of pixels	Pixel equivalent
1	30.02	882.26	0.034026
2	30.00	882.10	0.034010
3	30.00	882.11	0.034009
4	30.01	882.20	0.034017
5	30.02	883.01	0.034000



Fig. 13. Hardware platform of the online visual measurement system.

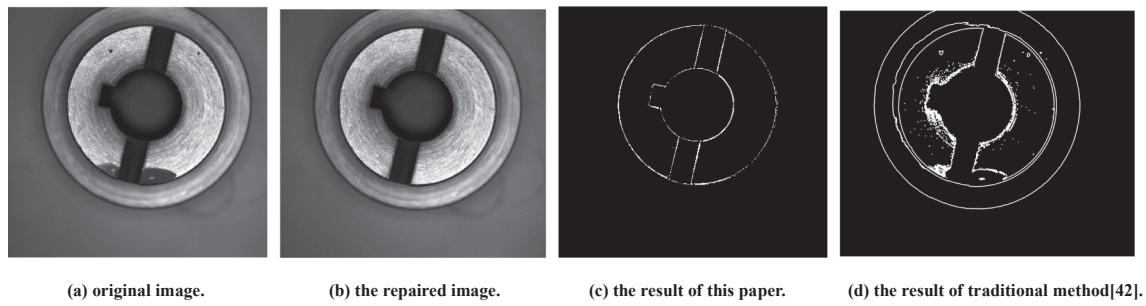


Fig. 14. The results of experimental image.

Table 7

The measurement results of the workpiece outer circle diameter.

Number of measurements	The result of CMM	The result of traditional method [42]	The result of this paper	Absolute error	Relative error
1	44.89		44.866	0.024	0.053%
2			44.910	0.020	0.045%
3		–	44.868	0.022	0.049%
4			44.867	0.023	0.051%
5			44.912	0.022	0.049%

the actual dimension of a pixel on the image.

A standard block of 30 mm was measured five times and the result of measurements is shown in Table 6. The average value of multiple measurements is 0.034012 mm/pixel, which will be used in subsequent experiments under the same physical environment.

After the calibration, the workpiece can be measured automatically. Firstly, set internal calibration parameters of the camera and pixel equivalent. Then, the camera will acquire the workpiece images after the CNC machine tool finished the current program. Finally, the workpiece images are processed by a series of methods mentioned above, the dimension of the workpiece is measured according to the pixel equivalent.

5.3. Analysis of experimental results

As shown in the Fig. 14(a), there are two pieces of cutting fluids and two pieces of chips on the workpiece. The dimension of the shaft is measured by the methods presented in this paper. After the detection and processing of interference regions on the workpiece surface, the image is shown in Fig. 14(b). Both cutting fluids and chips on the workpiece surface have been removed. The repaired image does not leave obvious voids and the traces of restoration were slight. The image is detected by the HED model and post-processed to obtain a rough edge image as shown in Fig. 14(c). The edge image has a complete contour without noise edges such as textures. As shown in the Fig. 14(d), it is the result of traditional edge detection methods [42]. It is difficult or impossible to measure the outer circle of the workpiece.

Based on cubic spline function, the edges shown in Fig. 14(c) are interpolated. The obtained sub-pixel edge coordinates are fitted by the least square method to obtain the equation of circle, which are as follows:

$$(a - 801.897)^2 + (b - 1133.7)^2 = 659.561^2 \quad (3)$$

where the unit is the number of pixels.

The number of pixels of the workpiece outer circle diameter was obtained by five repeated experiments and was converted into the actual dimensions by the pixel equivalent. It is considered that the result of CMM is the true value and the result of this paper is the observed value. The following two formulas (4)–(5) are used to calculate absolute and

relative errors, and the results are listed in Table 7.

$$\Delta = x - \mu \quad (4)$$

$$\delta = \frac{\Delta}{\mu} = \frac{x - \mu}{\mu} \quad (5)$$

where x is an observed value, μ is the true value, Δ is the absolute error, δ is the relative error.

As shown in the Table 7, the maximum absolute error of the diameter of the shaft is 0.024 mm and the relative error is 0.053%. The result meets the requirement of semi-finishing.

6. Discussion and conclusion

In this paper, an intelligent online visual measurement system based on deep learning was developed to solve problems of interference regions processing and edge detection in industrial measurement.

According to the above, the method in this paper has achieved good results in each stage. Compared with FMM and NS methods, the method selected in this paper can remove the interference regions effectively. Compared with the traditional edge detection method, the improved HED method can effectively detect the edge of the workpiece and achieve the measurement accuracy of 0.02.

Deep learning has been successfully applied to the field of industrial measurement. In addition, it provides methodological guidance for the application of deep learning in industrial measurement. Future work will continue to explore new solutions based on this method, expand the scope of application, reduce errors and improve measurement accuracy. The detection of the surface finish quality will also be studied in the future.

CRediT authorship contribution statement

Xuebing Li: Formal analysis, Investigation, Writing – original draft. **Yan Yang:** Software, Investigation. **Yingxin Ye:** Supervision, Writing – review & editing. **Songhua Ma:** Project administration. **Tianliang Hu:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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